

Customer Churn Prediction & Risk Segmentation Dashboard

Business Report & Findings Summary

[Telco Customer Churn Dataset]

Executive Summary

This project presents an end-to-end Customer Churn Prediction pipeline built using the **Telco Customer Churn dataset containing 7,043 customer records and 21 features**. The primary objective of the project was to identify customers who are likely to discontinue telecom services, understand the major factors contributing to churn, and provide actionable business recommendations to reduce customer loss.

The project involved **data preprocessing, exploratory data analysis, feature engineering, machine learning model training, risk segmentation, and business insight generation**. Three machine learning models were trained and compared: Logistic Regression, Random Forest Classifier, and Gradient Boosting Classifier.

After evaluation using Accuracy, Precision, Recall, F1-Score, and ROC-AUC metrics, **Logistic Regression** emerged as the best-performing model with the highest ROC-AUC score of **0.7135**. Since the dataset is imbalanced, ROC-AUC was considered the most reliable evaluation metric.

The best model was further used to segment customers into **High Risk, Medium Risk, and Low Risk churn** categories using churn probability scores.

Dataset Overview

Attribute	Value
Total Customers	7,043
Total Features	21
Churn Rate	~30% Yes, ~70% No
Train/Test Split	80% / 20%

Target Column	Churn
New Features Created	Charges Per Month, Senior With No Support, Average monthly spending based on active services,New Customer

Data Preprocessing & Feature Engineering

The dataset was cleaned and preprocessed before training the models. Missing values in the Total Charges column were handled carefully by converting the column into numeric datatype and replacing invalid values using **median imputation**.

Categorical features were encoded using One-Hot Encoding because the dataset contained many nominal categorical variables such as Contract, Internet Service, and Payment Method. One-Hot Encoding prevents incorrect ordinal relationships between categories.

Two new features were engineered:

- 1. Charges Per Month = Total Charges / tenure
- 2. Senior With No Support = Senior Citizen customers without TechSupport

Numerical features were standardized using StandardScaler before model training.

Model Comparison Results

Metric	Logistic Regression	Random Forest	Gradient Boosting
Accuracy	0.7197	0.7012	0.7012
Precision	0.5714	0.5067	0.5072
Recall	0.2736	0.2665	0.2476
F1-Score	0.3700	0.3493	0.3328
ROC-AUC	0.7135	0.6848	0.6971

Logistic Regression achieved the best ROC-AUC score and overall balanced performance. Therefore, it was selected as the final model for customer risk prediction.

Customer Risk Segmentation

Using the predicted churn probabilities from the best model, customers were grouped into three risk categories:

Risk Tier	Probability Threshold	Customer Count	Avg Monthly Charges	Avg Tenure
High Risk	>= 0.70	4	\$105	5.3 months
Medium Risk	0.40 – 0.69	403	\$80	25.4 months
Low Risk	< 0.40	1,002	\$60	39.6 months

This segmentation helps the business identify customers who require immediate retention strategies.

Key Insights

Top Factors Driving Churn

1. Contract Type — Customers on month-to-month contracts showed significantly higher churn rates compared to one-year and two-year contract customers.
 2. Tenure — Customers with lower tenure were more likely to churn, especially within the first 12 months.
 3. Monthly Charges — Customers paying higher monthly charges demonstrated increased churn probability.
 4. Payment Method — Electronic check users showed higher churn compared to customers using automatic payment systems.
 5. Lack of Add-On Services — Customers without tech support, online security, or backup services were more likely to leave.
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High Risk Customer Profile

High-risk customers generally shared the following characteristics:

- Month-to-month contract type
- Short tenure period
- Higher monthly charges
- Lack of support services such as TechSupport and OnlineSecurity
- Fiber optic internet users with premium billing

These patterns indicate that newer customers paying higher charges without long-term commitment are the most vulnerable segment.

Business Recommendations

Recommendation 1 — Launch an "Annual Plan Conversion" Campaign

Proactively reach out to month-to-month customers in their first 3 months with a personalized offer: 10–15% discount if they switch to a 1-year plan. This single action can reduce churn probability dramatically. Priority = customers paying over \$70/month.

Recommendation 2 — Build a "First 90 Days" Onboarding Program

Since churn peaks in the first year, invest in early customer success: welcome call at Day 1, check-in at Day 30, free trial of tech support, loyalty reward at 6 months. Preventing early churn is 5–10x cheaper than trying to win customers back.

Model Limitations & Future Improvements

- **Low Recall (~27%):** We miss many churners. Could improve by adjusting the decision threshold

- **No time-series data:** If we had logs showing declining usage, increasing support calls, etc., accuracy would improve significantly.
 - **Model drift:** Customer behavior changes over time. Retrain the model every quarter with fresh data.
 - **External data missing:** Competitor pricing, regional outages, and promotions aren't captured here.
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Conclusion

This project successfully demonstrated how machine learning can be used to predict customer churn and generate business-driven insights. Logistic Regression proved to be the most effective model based on ROC-AUC performance. The analysis revealed that contract type, tenure, and monthly charges are the strongest indicators of churn.

The customer risk segmentation approach provides valuable support for proactive retention strategies, helping businesses reduce revenue loss and improve customer satisfaction.
